

2024

(Session : 2024-26)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

1. Answer any **four** questions of the following :

$$3\frac{1}{2} \times 4 = 14$$

- (a) What is D' Alembert's principle ?
- (b) What are infinitesimal generators ?
- (c) Define Poisson's bracket.
- (d) Define Dirac-Delta function.
- (e) Find the commutation relation between x and p .

(f) Define angular momentum operator ?

Section – B

Answer any **four** questions of the following :

2. Deduce Hamilton's equation and give their applications. 8+6 = 14
3. State and prove Hamilton-Jacobi equation. 14
4. State and prove Poisson's theorems. 14
5. Find the eigenvalue and eigenfunction of one-dimensional harmonic oscillator using Schrodinger equation. 14
6. Prove the following commutation relations :
 - (a) $[L_x, L_y] = i\hbar L_z$ 7
 - (b) $[J_+, J_-] = 2\hbar J_z$ 7
7. Discuss addition of angular momenta and find out the Clebsch-Gordon coefficients. 14
8. Discuss operator formalism in quantum mechanics. Define momentum operator.

8+4 = 14

9. Write notes on any **two** of the following :

$$7+7 = 14$$

- (a) Generating functions
- (b) Canonical transformations
- (c) Hermitian operators and their properties
- (d) Schrodinger Picture

